

Constitutional AIOps - All Benchmark Tables

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Table 1: Comprehensive Benchmark Results

Task	Agent	N	Accuracy	BERT-F1	Cos Sim	Term Overlap	P50 (ms)	P95 (ms)
Annotation	qwen3:4b-instruct	100	89.0%	0.516	0.265	0.178	1872	2284
RCA	qwen3:14b	50	94.0%	0.341	0.387	0.733	14075	22718
Overall	Hybrid	150	90.7%	0.458	0.305	0.411	2100	17374

Table 2: Per-Source Breakdown

Task	Source	N	Accuracy	BERT-F1	Cos Sim
Annotation	Loghub Bgl	28	71.4%	0.503	0.223
Annotation	Loghub Hdfs	72	95.8%	0.521	0.282
Rca	Lemma Rca Cloud	28	100.0%	0.353	0.480
Rca	Opseval 5G Communication	2	100.0%	0.302	0.282
Rca	Opseval Mobile Communication Network	4	100.0%	0.353	0.334
Rca	Opseval Wired Network	16	81.2%	0.323	0.249

Table 3: Error Analysis

Failure Mode	Count	Description
BGL False Positive	8	ANN_118, ANN_142, ANN_141, ANN_135, ANN_119 (+3 more)
ANNOTATION Incorrect (loghub_hdfs)	3	ANN_049, ANN_034, ANN_012
RCA Incorrect (opseval_Wired Network)	3	RCA_060, RCA_040, RCA_039
Total Failures	14	14/150 = 9.3% error rate

Table 4: Ablation Study - Component Contributions

Configuration	Ann Acc	RCA Acc	Overall	BERT-F1	Cos Sim	Term Ov.	Avg Latency	Delta vs Full
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Configuration	Ann Acc	RCA Acc	Overall	BERT- F1	Cos Sim	Term Ov.	Avg Latency	Delta vs Full
Full System (Hybrid dual-agent baseline)	89.0%	98.0%	92.0%	0.459	0.308	0.433	5917ms	-
Single Agent - Qwen3-4B for both annotation and RCA	89.0%	94.0%	90.7%	0.457	0.309	0.401	6033ms	-1.3%
Single Agent - Qwen3-14B for both annotation and RCA	89.0%	88.0%	88.7%	0.458	0.306	0.427	5988ms	-3.3%
No Structured Output - raw text, no JSON metadata extraction	89.0%	96.0%	91.3%	0.457	0.302	0.383	6157ms	-0.7%
No System Prompt - empty system message, raw user query only	45.0%	92.0%	60.7%	0.389	0.288	0.780	11018ms	-31.3%
With Graph Context - historical episode context injected (RAG)	89.0%	90.0%	89.3%	0.454	0.288	0.335	6430ms	-2.7%
No Constitutional AI - skip validation (overhead measurement)	89.0%	96.0%	91.3%	0.457	0.307	0.420	6110ms	-0.7%

N=150 per configuration. Dataset: curated_150 (seed=42). Temperature=0.0.

All values from actual benchmark runs. No fabricated or estimated data.